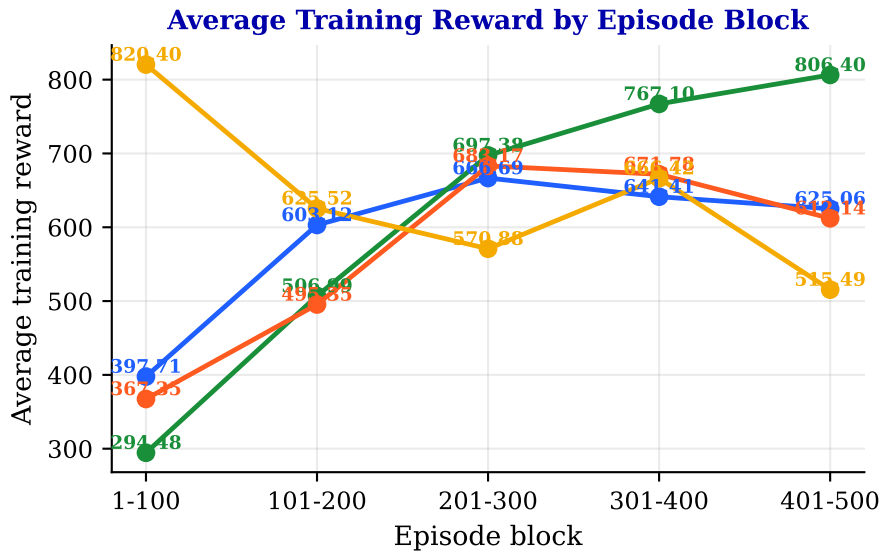


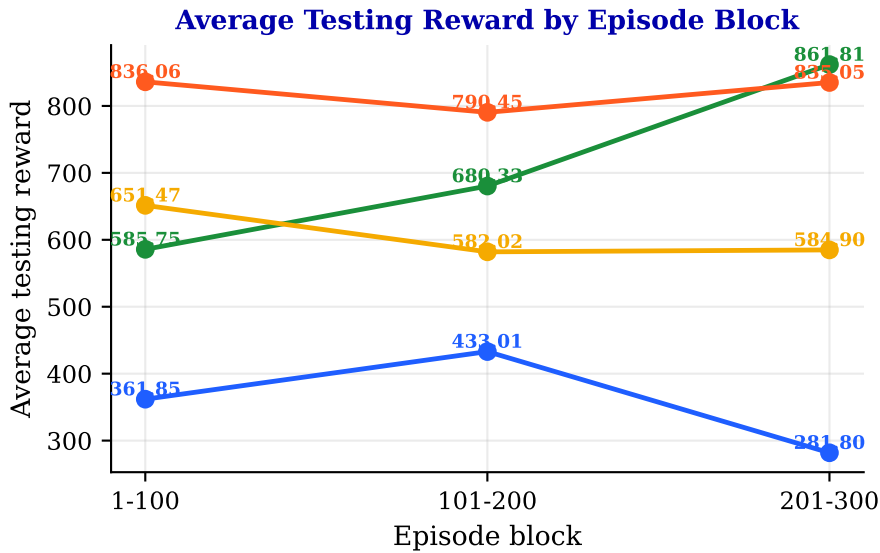
DQN in CARLA - Training & Testing Dashboard (LR multiplier = 1, Final LR = 5e-05)

All rewards are ENVIRONMENT rewards only • Training Episodes: 500 • Testing Episodes: 300 • Episode Block Size: 100 • Epsilon: 0.2



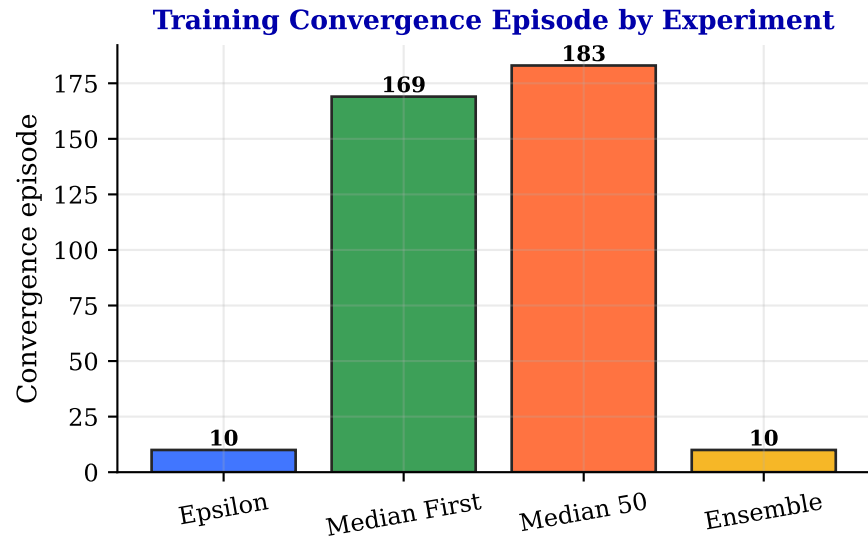
Training Reward Summary Statistics (All 500 Episodes)

Method	Mean	Median	Std	Min	Max	Q1	Q3
Epsilon	586.80	656.27	435.09	-1569.51	1219.17	229.70	959.63
Median First	614.47	743.82	462.58	-1578.61	1199.47	275.46	1003.68
Median 50	565.96	586.97	411.29	-978.22	1164.75	131.49	946.94
Ensemble	639.74	671.53	563.98	-1238.55	1286.28	107.55	1208.47



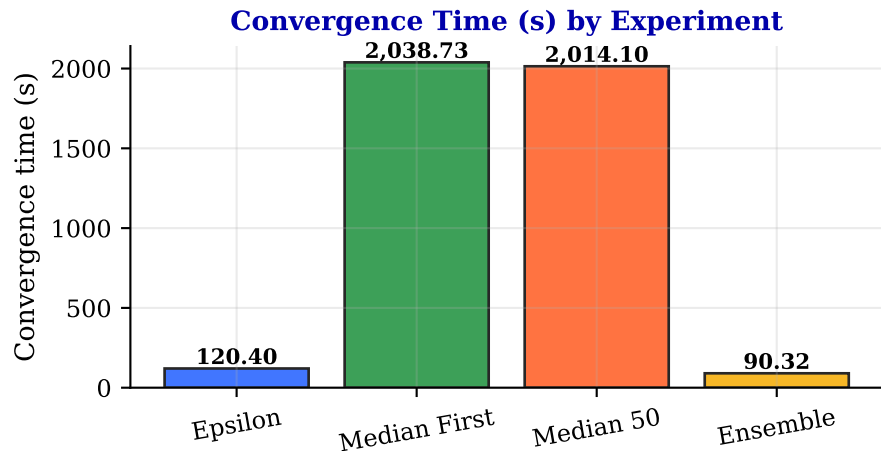
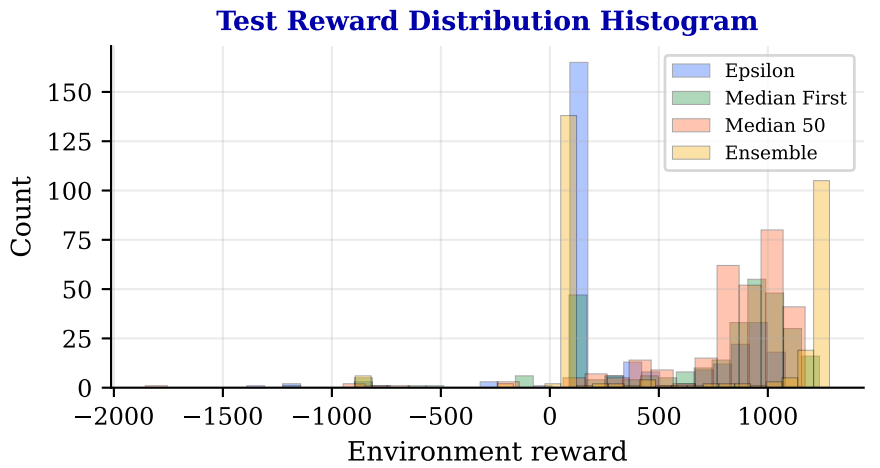
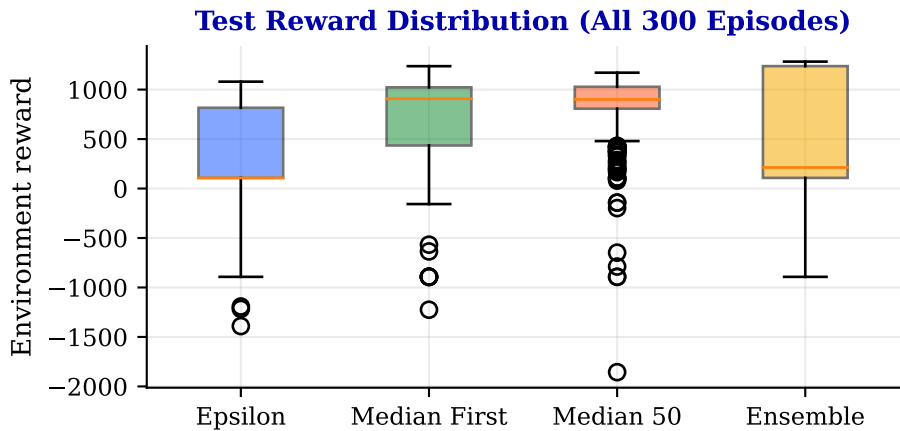
Testing Reward Summary Statistics (All 300 Episodes)

Method	Mean	Median	Std	Min	Max	Q1	Q3
Epsilon	358.89	107.55	430.92	-1390.32	1080.12	107.55	816.19
Median First	709.30	907.16	454.72	-1224.80	1236.55	435.39	1022.44
Median 50	820.52	900.67	355.75	-1855.94	1170.85	806.94	1028.03
Ensemble	606.13	212.96	583.66	-892.45	1282.66	107.55	1236.86



Convergence Summary

Method	Conv Episode	Train Episodes	Train Time(s)	Conv Time(s)
Epsilon	10	500	6,020.03	120.40
Median First	169	500	6,031.75	2,038.73
Median 50	183	500	5,503.00	2,014.10
Ensemble	10	500	4,515.91	90.32



Key Takeaways

- Median 50 has the highest average test reward (820.52).
- Ensemble has the fastest convergence time (90.32s).
- Median 50 has the lowest test reward std (355.75).
- Test networks are frozen; no updates occur during testing.

Best Overall

MEDIAN 50

Average Test Reward: 820.52
Convergence Time: 2014.10s

Ranking uses reward rank + convergence-time rank.

Experiment Settings

Environment : CARLA 0.9.14 (Real)
Algorithm : DQN
Network : RND + Count + Target + Replay + Adam
NoisyNet : Removed
Learning Rate Mult. : 1
Final Learning Rate : 5e-05
Epsilon : 0.2
Training Episodes : 500
Testing Episodes : 300
Episode Block Size : 100